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A Report On

Land Use and Land Cover Change in Gandaki Province from 2011 to 2023

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1 Background

Land use and land cover change is one of the major areas of GIS analysis, which is used for various purposes. This analysis was done as a practice work to learn the raster analysis using GIS (see [1]).

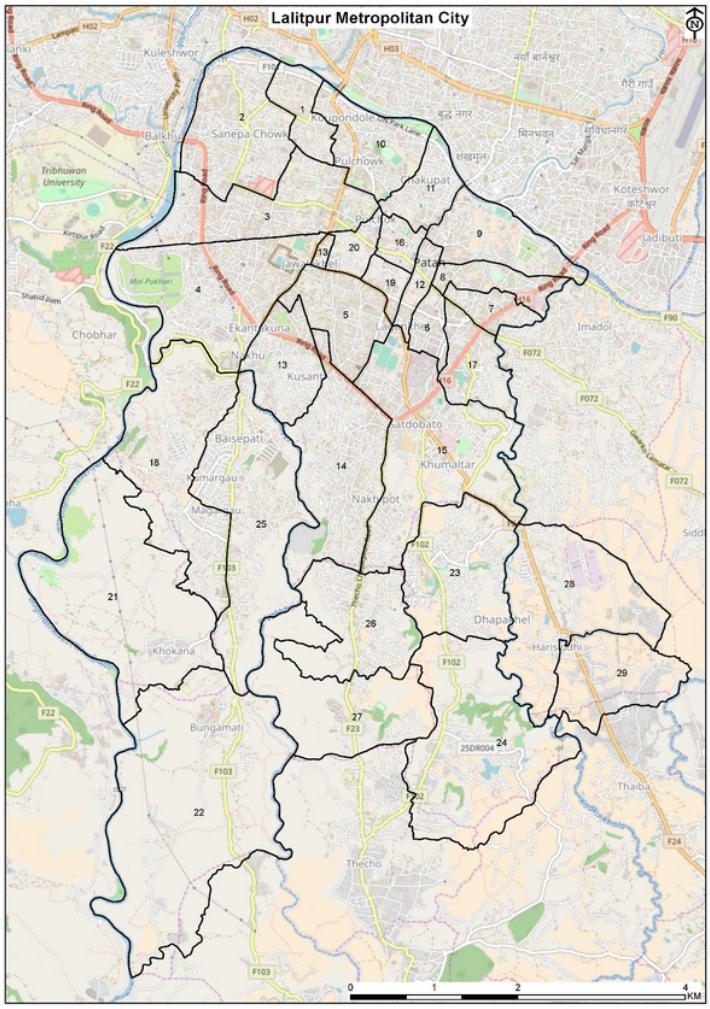

Figure 1.1: Main Web-GIS interface of the system

1.1 Data and Output

In this analysis MODIS land surface reflectance data which is provided by USGS freely was used. The data for four different years but the same date was used to understand the land use change in similar season for last one decade. The detail of the data used is listed below.

- Month: October first week
- Year: 2011, 2015, 2019, 2023

- Vector data of Province-4 (Gandaki) and its districts

NDVI (Normalized Difference Vegetation Index) was produced for each year and compared using graph.

1.1.1 MODIS Land surface Reflectance

MODIS (Moderate Resolution Imaging Spectroradiometer) is a satellite-based sensor used for earth and climate measurements aboard the Terra (originally known as EOS AM-1) and Aqua (originally known as EOS PM-1) satellites.([2]) MOD09 (MODIS Surface Reflectance) is a seven-band product computed from the MODIS Level 1B land bands 1 (620-670 nm), 2 (841-876 nm), 3 (459 479), 4 (545-565 nm), 5 (1230-1250 nm), 6 (1628-1652 nm), and 7 (2105-2155 nm). The product is an estimate of the surface spectral reflectance for each band as it would have been measured at ground level as if there were no atmospheric scattering or absorption. It corrects for the effects of atmospheric gases and aerosols.([3]) MOD09 data of any place on earth can be directly downloaded from US Geological survey (USGS) website of NASA, provided that we prompt the boundary of the place under study and the time interval. HDF (Hierarchical Data Format) files are downloaded from the website, bands 01 and 02 are imported into GIS under the default projection WGS 84 EPSG:4326.

1.1.2 NDVI

The Normalized Difference Vegetation Index(NDVI) is a widely-used metric for quantifying the health and density of vegetation using sensor data. It is calculated from spectrometric data at two specific bands: red and near-infrared. Healthy vegetation (chlorophyll) reflects more near-infrared (NIR) and green light compared to other wavelengths. But it absorbs more red and blue light.

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)}$$

The result of this formula generates a value between -1 and +1. If you have low reflectance (or low values) in the red channel and high reflectance in the NIR channel, this will yield a high NDVI value. And vice versa. Overall, NDVI is a standardized way to measure healthy vegetation. When you have high NDVI values, you have healthier vegetation. When you have low NDVI, you have less or no vegetation.([4])

- <0 : Snow, river
- 0-0.1 : River or Riverine sediments
- 0.1-0.3 : No vegetation, bare land
- 0.3-0.5 : Low vegetation, built up area
- 0.5-0.7 : Good vegetation
- 0.7 : Dense vegetation

2 Methodology

QGIS and Grass GIS were alternately used for this project.

2.1 QGIS Steps for analysis

1. Download data
    - Extract vector polygon data of the districts of Gandaki Province
    - Create KML file of the extent of data
    - Search for the data at `https://search.earthdata.usgs.gov` using key word **MOD09Q1** and the KML file for the extent
    - Download the data for the years 2011, 2015, 2019 and 2023 for every first week of October in the respective years
2. Clip and Transform data (Clip the data to the extent of the province and transform to WGS84-GCS(EPSG:4326) using QGIS

2.1.1 Clipping and re-projecting the MODIS Data

- Since a single raster layer covered our entire area of study, there was no need of merging more than one raster layer
- Raster clipping by mask layer was done. **MOD09** layers were kept as input later and district vector layer as mask layer
- Target CRS was set to WGS84-GCS(EPSG:4326)
- Layer was exported as .tiff file and saved into the provincial geopackage

2.2 GRASS Steps for analysis

1. Import the clipped data of both the bands for each year
2. Reclassify the raster data according to the color bands and Calculate NDVI
3. Perform a statistical analysis on the land use on each allotted time period
4. Go back to QGIS and create Graph from the land use comparison

2.2.1 Creating Location in Grass GIS

A Location contains Mapsets and each Mapset contains data related to a specific task, user or a smaller project. Given information are gathered to create a location in Grass GIS:

- coordinate system for your project area (plane or with projection and ellipsoid)
- minimum and maximum coordinates of the area of interest
- ground resolution

In order to create a new location, enter a non-existing location name, and GRASS will ask you if you would like to create a new location.

2.2.2 Importing raster and vector data

Following data were imported to Grass.

- Clipped and projected Modis data of each band (Red and NIR)
- Vector data of Districts

2.2.3 Calculation of NDVI

Following Grass GIS commands were used to calculate NDVI for each of the selected years:

```
- i.vi output=ndvi.2011 viname=ndvi red=clip2011_b1 nir=clip2011_b2
- i.vi output=ndvi.2015 viname=ndvi red=clip2015_b1 nir=clip2015_b2
- i.vi output=ndvi.2019 viname=ndvi red=clip2019_b1 nir=clip2019_b2
- i.vi output=ndvi.2023 viname=ndvi red=clip2023_b1 nir=clip2023_b2
```

2.2.4 Reclassification of NDVI

Firstly, a .txt file was formed with the following syntaxes for the re-coding process:

```
-1:0:0
0:0.1:1
0.1:0.3:2
0.3:0.5:3
0.5:0.7:4
0.7:1.0:5
```

Then, following Grass GIS commands were used to reclassify the NDVI for each of the selected years:

```
- r.recode input=ndvi.2011 output=reclass.2011 rules=C:\ndvirule.txt
- r.recode input=ndvi.2015 output=reclass.2015 rules=C:\ndvirule.txt
- r.recode input=ndvi.2019 output=reclass.2019 rules=C:\ndvirule.txt
- r.recode input=ndvi.2023 output=reclass.2023 rules=C:\ndvirule.txt
```

2.2.5 Conversion of district data from vector to raster

The imported district vector layer contains information on area of every districts within the province. Before the comparison process, the vector layer needs to be converted into raster where every districts has a unique identification based on their land area. Following Grass GIS commands were used to convert the vector to raster layer:

```
- v.to.rast input=District type=area output=Dist-rast use=attr
  ↪ attribute_column=cat label_column=district
```

2.2.6 Statistics of NDVI with respect to Districts

Following Grass GIS commands were used for the statistical comparison of ndvi and district area:

```
- r.stats -a -l -N input=Dist-rast, reclass.2011 output=C:\stats2011.
  ↪ csv separator=comma
- r.stats -a -l -N input=Dist-rast, reclass.2015 output=C:\stats2015.
  ↪ csv separator=comma
- r.stats -a -l -N input=Dist-rast, reclass.2019 output=C:\stats2019.
  ↪ csv separator=comma
- r.stats -a -l -N input=Dist-rast, reclass.2023 output=C:\stats2023.
  ↪ csv separator=comma
```

All the .csv files are compiled manually in a separate spreadsheet containing fields of information on land use over the allotted years.

2.2.7 Creation of Graphs

- Import the compiled spreadsheet in QGIS
- Open DB Manager in QGIS and open the Query window
- Merge the fields given in two attribute tables, one of the district vector layer and another of the compiled spreadsheet of land use statistics using the following SQL Code:

```
- select fid, district, area2011, area2015, area2019,
  ↪ area2023, area, t1.geometry from district t1, "stat
  ↪ all" t2
where t1.district = t2."Name of District"
and t2.lu_id=0
```

- Load the layer for each land use ID from 0 to 5
- Update the layer properties to display histogram of land use in the year 2011, 2015, 2019 and 2023

3 Result of Analysis

The histogram comparing the land use in year 2011, 2015, 2019 and 2023 in each districts of Gandaki Pradesh was obtained. This has been illustrated in the atlas maps below:

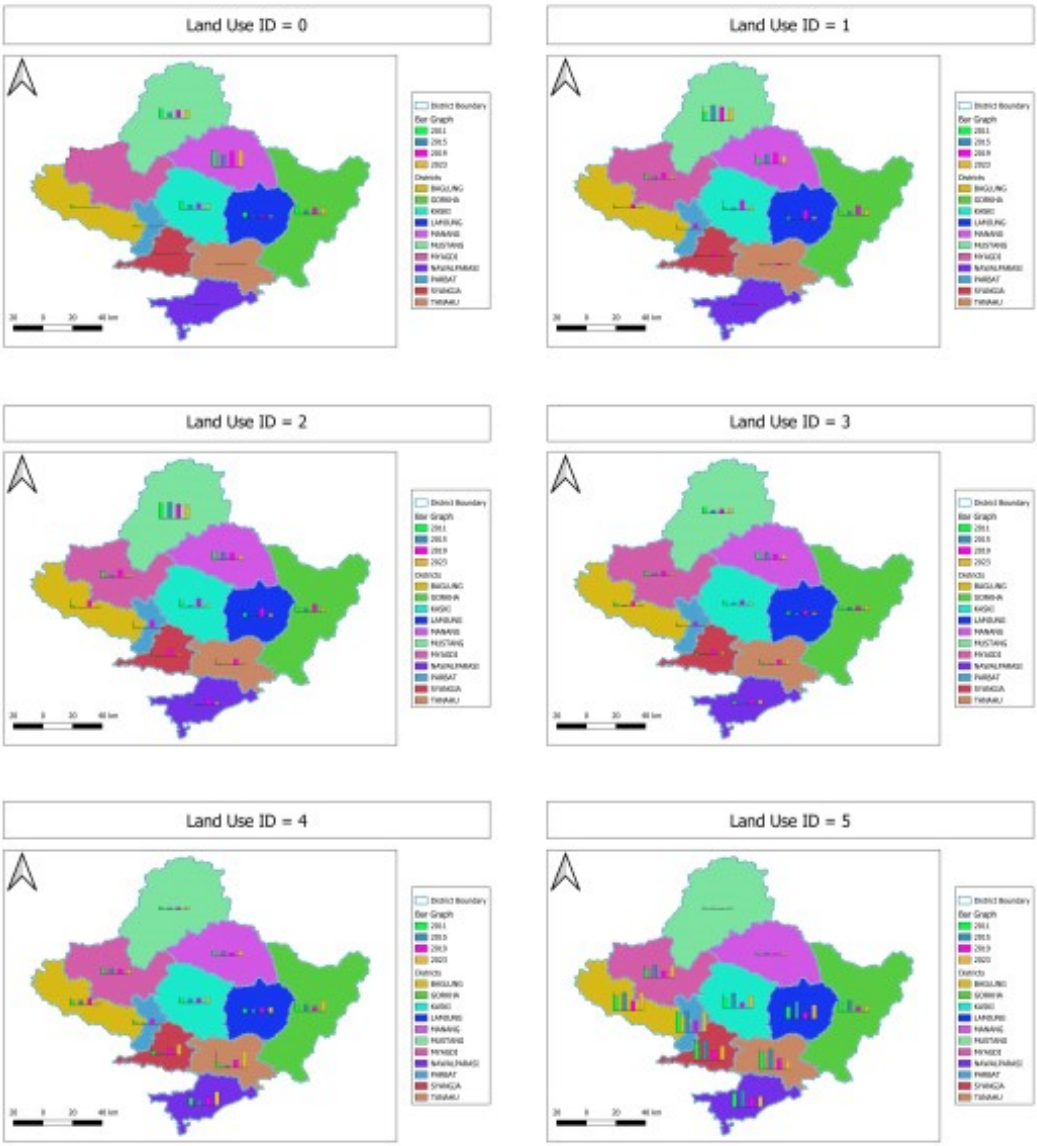

Figure 3.1: Histogram for each district's land use over the years

Following graphs illustrate the total area occupied by the districts in each NDVI case in the given years.

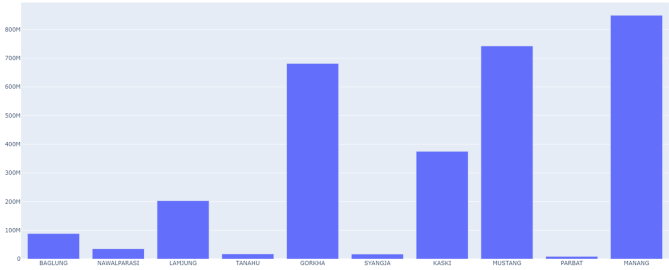

Figure 3.2: NDVI<0 in 2011

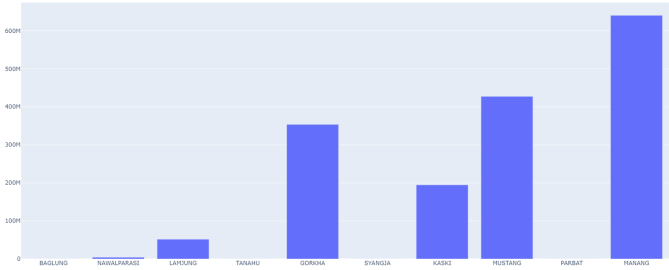

Figure 3.3: NDVI<0 in 2015

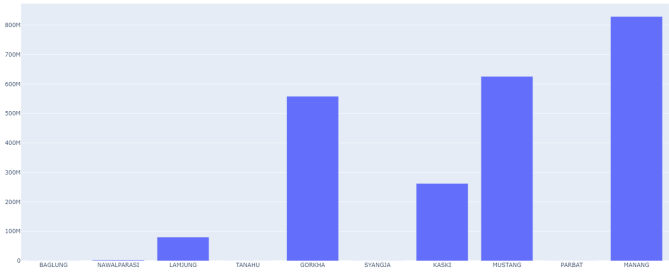

Figure 3.4: NDVI<0 in 2019

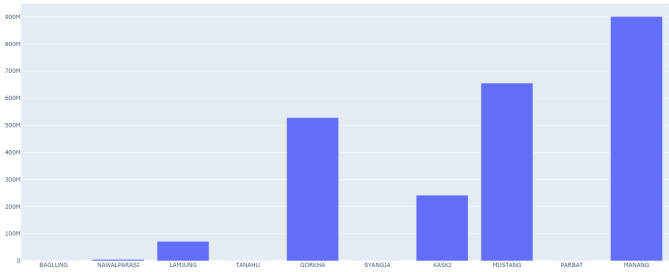

Figure 3.5: NDVI<0 in 2023

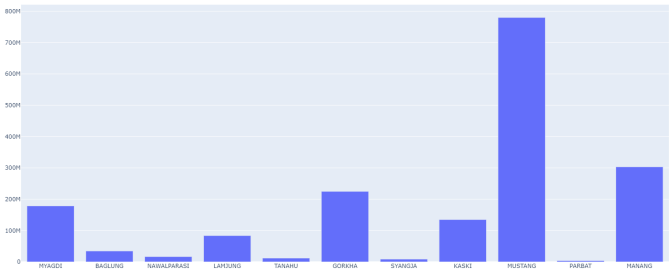

Figure 3.6: NDVI:0-0.1 in 2011

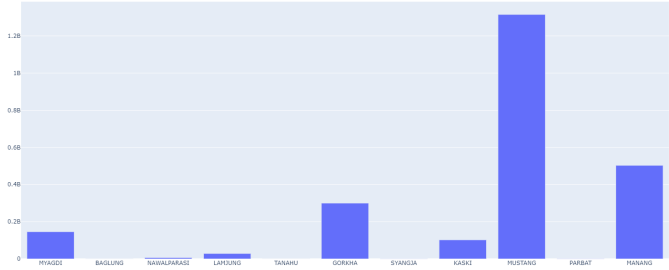

Figure 3.7: NDVI:0-0.1 in 2015

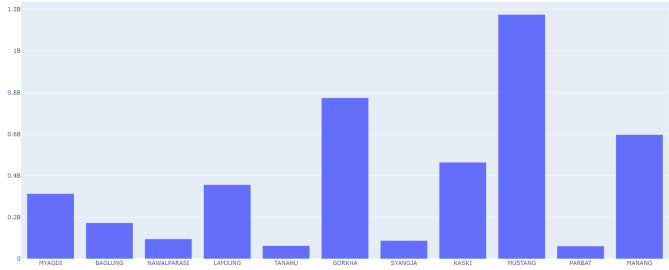

Figure 3.8: NDVI:0-0.1 in 2019

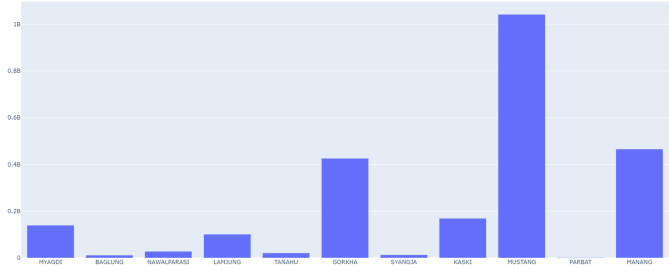

Figure 3.9: NDVI:0-0.1 in 2023

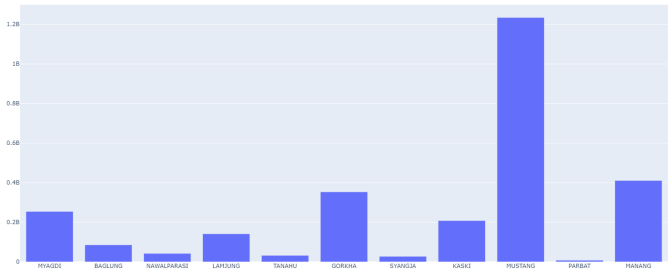

Figure 3.10: NDVI:0.1-0.3 in 2011

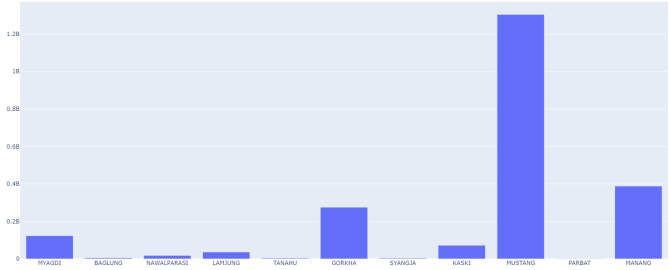

Figure 3.11: NDVI:0.1-0.3 in 2015

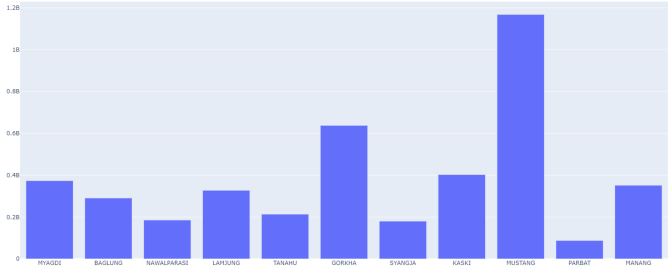

Figure 3.12: NDVI:0.1-0.3 in 2019

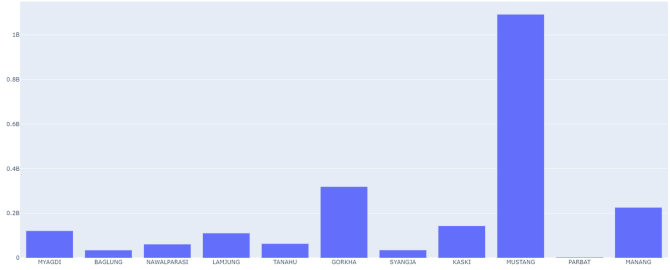

Figure 3.13: NDVI:0.1-0.3 in 2023

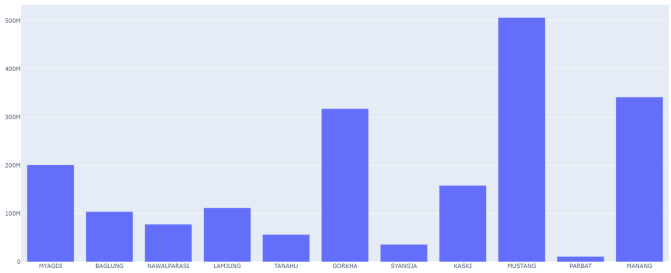

Figure 3.14: NDVI:0.3-0.5 in 2011

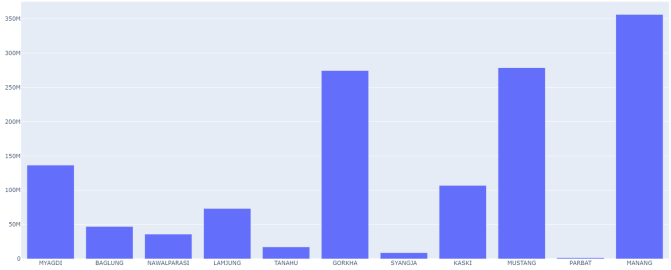

Figure 3.15: NDVI:0.3-0.5 in 2015

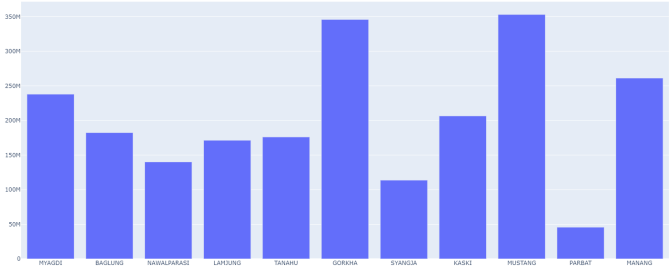

Figure 3.16: NDVI:0.3-0.5 in 2019

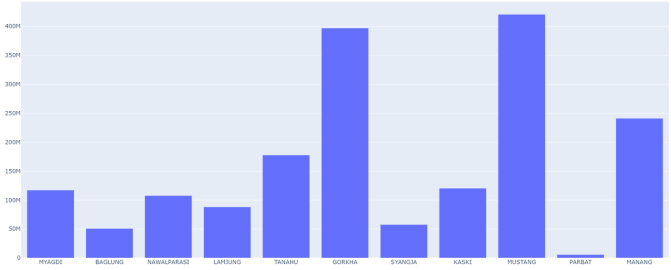

Figure 3.17: NDVI:0.3-0.5 in 2023

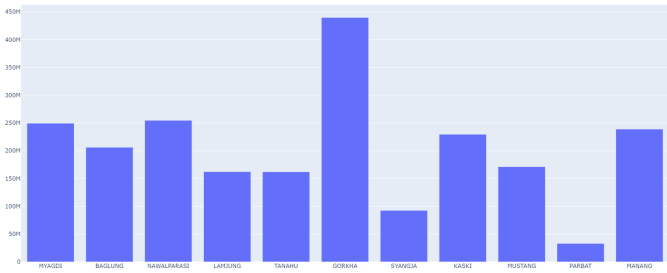

Figure 3.18: NDVI:0.5-0.7 in 2011

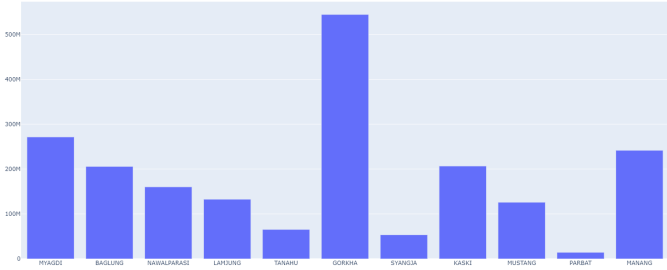

Figure 3.19: NDVI:0.5-0.7 in 2015

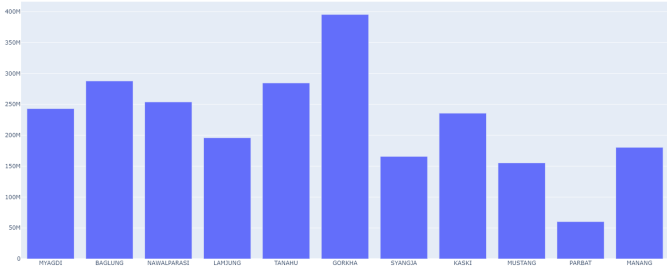

Figure 3.20: NDVI:0.5-0.7 in 2019

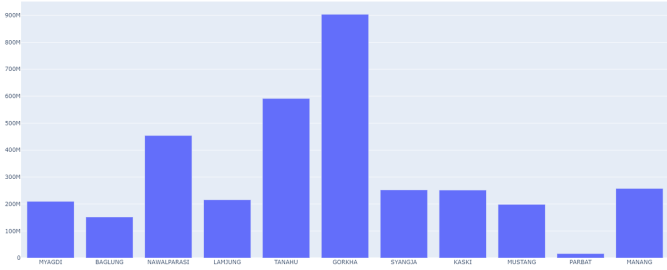

Figure 3.21: NDVI:0.5-0.7 in 2023

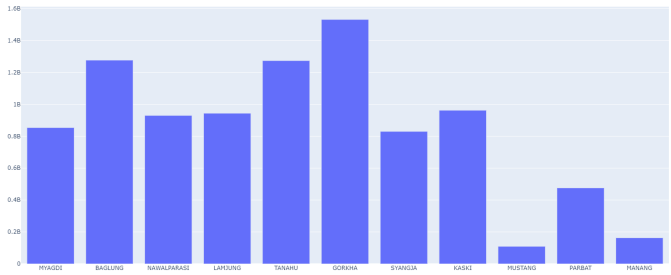

Figure 3.22: NDVI>0.7 in 2011

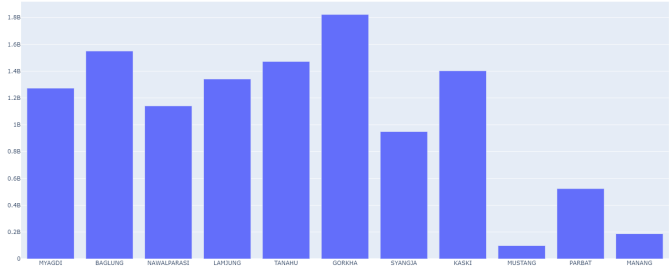

Figure 3.23: NDVI>0.7 in 2015

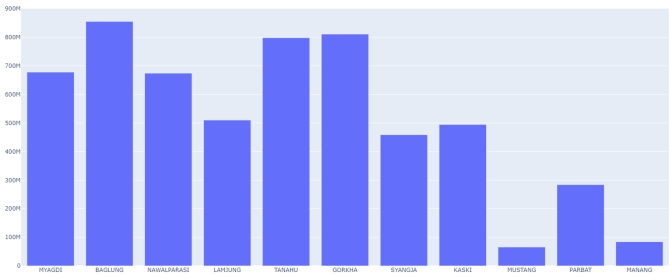

Figure 3.24: NDVI>0.7 in 2019

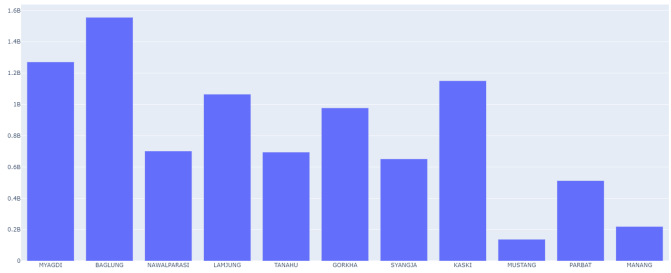

Figure 3.25: NDVI>0.7 in 2023

4 Conclusion and Discussion

From the bar graphs and histograms, we can conclude the following:

- Area with low to no vegetation are found as we move up north towards Manang and Mustang and healthier vegetation is found down south in the hilly region and terai region
- Built up area has been of the same percentage through out the years in the districts
- Healthier vegetation was in a decline in all districts in 2019 but that condition has been reverted in 2023
- Manang, Mustang and Gorkha has the greatest land area with river and snow
- Gorkha, Tanahu and Kaski have had good vegetative area throughout the years

References

[1] P. Devkota, S. Dhakal, S. Shrestha, and U. B. Shrestha, "Land use land cover changes in the major cities of nepal from 1990 to 2020," *Environmental and Sustainability Indicators*, vol. 17, Feb. 2023.

[2] *Terra and aqua*, Accessed: 2024-2-15, 2024. [Online]. Available: `https://modis.gsfc.nasa.gov/about`.

[3] *Surface reflectance*, Accessed: 2024-2-15, 2024. [Online]. Available: `https://modis-land.gsfc.nasa.gov/surfrad.html`.

[4] *Ndvi overview*, Accessed: 2024-2-15, 2024. [Online]. Available: `https://gisgeography.com/ndvi-normalized-difference-vegetation-index/`.